

# Planning Fast and Slow: Guided Trajectory Synthesis via Conditional Flow Matching

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## WORD COUNT

Number of Pages:

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## ABSTRACT

This should provide a summary of what highlights the reader is going to expect to read. The length should be short, 200-300 words. An abstract should include: The significance of the work, which may require a brief overview of the topic area. The contributions that this work is making to the topic area, the key results that are presented Achievements to date, what they were and how they are paving the path for future work.

Note that this is the first thing that the examiner is going to read possibly after taking a glance at the title. They need to be informed about what they are going to expect to find in this report.

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## 1 INTRODUCTION

Imbuing intelligent agents with the capability for autonomous-decision-making in real-world environments is challenging. For decades, reinforcement learning has offered a robust mathematical framework for acquiring optimal behavioral skills through interactions within an environment to maximize cumulative rewards (Sutton and Barto 2014). The integration of deep neural networks as high-capacity function approximators has propelled RL even farther, leading to remarkable achievements across diverse domains, from mastering complex games to enabling basic robotic locomotion and manipulation (Chen et al. 2018).

However, the widespread success of modern machine learning, as seen in areas such as computer vision and natural language processing, fundamentally hinges on the availability of large and highly diverse datasets. In the conventional RL paradigm, the acquisition of such extensive data becomes prohibitively impractical for real-world applications. For example, autonomous driving would need to practice in numerous cities for every model update, which is costly, time-consuming, let alone being unsafe. This inherent limitation created a gulf of generalization between lab-bound RL successes and the complex, open-world settings where intelligent decision-making is most critically needed.

Offline Reinforcement Learning emerges as a pivotal paradigm to bridge this gap. Offline RL focuses on enabling the training of policies solely from previously collected datasets, without additional online interaction. This approach holds immense potential for transforming vast amounts of historical data into robust, generalizable decision-making engines, much like how supervised learning revolutionized pattern recognition. The implications for fields such as robotics, healthcare, and autonomous driving are profound, as it offers a viable pathway to deploy sophisticated AI systems where active data collection is either unfeasible or too costly. Beyond these, offline RL can revolutionize domains such as operations research, logistics, power plant control, inventory management, and finance, all of which have rich historical data where active exploration is difficult or expensive (Levine et al. 2020).

Finally, taking a page out of the rise of vision language models, the ability to leverage large and diverse datasets in an offline setting further enables the development of foundational Vision-Language-Action models or generalist robot models. This means that pre-training on massive, heterogeneous and potentially multi-modal datasets can lead to acquisition of broadly generalizable skills. These can then be efficiently fine-tuned for specific tasks in deployment. In a way, this paradigm allows bootstrapping on pre-trained visual language models to learn behaviors. Passive data can now be transformed into active intelligence by learned observations (Kim et al 2024).

### 1.1 Background and Context

Despite its immense promise, offline RL is plagued by algorithmic challenges. At its core, offline RL is fundamentally about making and answering counterfactual queries (Levine 2020). These are “what if” questions that involve forming hypotheses about what might happen if the agent were to carry out actions different from those observed in the training data. In other words, the ability to improve upon the behavior seen in the dataset, it must execute sequences of actions that diverge from the training data. This inherent need for counterfactual inference strains conventional machine learning tools, which are typically designed under the assumption that test-time queries come from the same distribution as the training data, independently and identically distributed.

Consequently, making such counterfactual queries results in distributional shift. When a learned policy attempts to take actions or visit states that deviate from those observed in the training data, conventional RL algorithms can produce erroneous and excessively optimistic Q-value estimate. This leads to policies that are brittle and perform poorly when deployed.

This issue is particularly exacerbated by the use of high-dimensional and expressive function approximators such as DNN. The errors can accumulate quadratically over long horizons and make stable learning difficult. Furthermore, real-world datasets, especially those derived from human demonstrations or varied sources, inherently exhibit multimodal action distributions. This means that there are often multiple valid ways to perform a given task or different sequences of actions to achieve a long-horizon goal. Traditional behavior cloning methods frequently struggle to capture this multimodality effectively, resulting in jerky, hesitating, or stuck motions and failing to generalize beyond narrowly

observed demonstrations. This inadvertently average across modes and leads to indecisive behaviors.

In recent years, generative modeling, particularly with denoising diffusion probabilistic models, has shown remarkable capability in generating high-quality and diverse images. These models learn by iteratively refining random noise into coherent output and offer significant advantages in addressing distributional shift when applied in robotics, especially in behavior cloning:

1. Diffusion models gracefully capture high-dimensional and multimodal action spaces, as evidenced by their success in image and video generation.
2. They exhibit impressive training stability, requiring little task-specific hyperparameter tuning compared to models like generative adversarial networks.
3. The most surprising property is temporal compositionality. The models can essentially “stitch together” in-distribution subsequences from training data to generalize to novel tasks.

This enables counterfactual queries because planning can now be framed as a sampling conditioned on desired outcomes. Just like in image generation, the iterative sampling process of diffusion models allows for flexible conditioning. This allows a single diffusion model to be used for multiple tasks in the same environment, even for tasks unseen during training.

## 1.2 Objectives

The objectives should be clearly listed here to explain what new outputs you plan to achieve in this work. After the abstract, this section is the one next most important thing your examiner will read. They want to know what you plan to do and see if you are on target to achieve it. You will have listed these objectives in your first report that you submitted at the start of your project. Therefore copy and paste them into this section and show them in bullet points each with no more than one sentence like below:

- Objective 1.....

- Objective 2 ....
- Objective 3 ....
- ..
- Objective N

You may find that you need to slightly revise your objectives since writing your first report. This is totally acceptable if you see a need but you are advised to discuss any revisions with your supervisor. Be very clear to understand what is an objective. An objective is a new output providing new knowledge or bringing knowledge together. Therefore statements like “learn about ...”, “perform simulations on...” or “write software code for...” are not objectives, they are tasks that would possibly be required to form an objective. Therefore objectives are not tasks; they are statements of what advances the work aims to or will make.

Having listed your objectives you may wish to summarise below the tasks you have carried out and will carry out to fulfil the objectives, so that for example it gives an idea of what work was involved in the project such as building new hardware, performing simulations or carrying out theoretical analysis.

### 1.3 Achievements

Summarise what you have achieved in your work, what are the highlights that the reader should look forward to reading about?

### 1.4 Overview of Dissertation

Briefly overview the contents of what follows in the dissertation explaining what is in each chapter and explain it in a way that show the narrative of the work so that it makes sense to the reader why the dissertation is organised in the way it is.

## 2 BACKGROUND THEORY AND LITERATURE REVIEW

This chapter is required to include a more deep literature review beyond what you briefly introduced in section 1.1. Furthermore it should include the background theory and concepts to what is critical to your work. It is very important to consider what sections you should form within this chapter and how to structure it. You should send a planned contents of this and other chapters to your supervisor before you write up and ensure they are happy with what you propose to do.

The earlier sections of this chapter should include a literature review of the state of the art in your topic area citing books, publications and possibly some other web sources of work relevant to the topic. They should be written up in a way that they lead the reader to realising where your contributions to the work are useful and filling a gap. Don't just assume the reader will know what you write about, as your second examiner is not considered to be expert in the area and will want to learn from what you write about.

There will be background theory and concepts that are directly relevant to your work. It is important to work out what these are and develop appropriate sections for this purpose. Again the reader may need to learn about these so it is important that you explain them clearly and in the context that is relevant to your work, not just repeating what all the other literature says. You need to show you understand it by explaining it.

You should ensure that at this point in the chapter a paragraph or two is presented to show what content will be in this chapter and this significance in order to indicate where it is going to lead the reader.

### 2.1 Section Title

Text for section

### 2.2 Section Title

Continue with as many sections as you need

Note that references work like this [3] [2]

## 2.3 Summary

Your final section should include a summary of the chapter, which will summarise what has been covered and where there are gaps in knowledge which pave the way into the original contributions you make in the coming chapters.

### 3 TECHNICAL CHAPTER

The technical body of the report consists of a number of chapters (just one here, but there will usually be more) of work you have carried out to date. Follow a logical structure in how you present your work. This will usually be the phases of the work you carried out which will link loosely to the objectives. Note that these chapters are not meant to be a log book, thus do not write them like a diary of work carried out but as a technical report specifying clearly what has been implemented and presented in a way that someone else could carry out the work based on what they have read.

Again you need to consult with your supervisor on the technical chapters you plan to write by providing a draft contents page. It should show clearly the sections in it and every chapter should have a clear paragraph introducing it and what to expect in it.

#### 3.1 First Section

Structure your text into sections.

##### 3.1.1 First Subsection

If necessary, also use subsections. Subsections are entered using the Heading 3 paragraph style (all these heading styles are self-numbering).

##### 3.1.1.1 First Subsubsection

If you really need subsubsections.

##### 3.1.2 Second Subsection

And, as required, more subsections.

#### 3.2 Second Section

As an example of a figure, consider Figure 3.1 and note how it is referenced here by cross referencing to a caption method so that also you can form a list of figures at the start. You should be familiar with how to insert a caption and refer to it in Word from the training needs analysis form you were given. The caption should include title and essential

information about figure and figures must always be referred to in the text. Ensure you insert figures into appropriate points in the text so they are both near to the text that is referring to the figure but also ensure that they are arranged in a tidy fashion so they do not leave big gaps with no text.

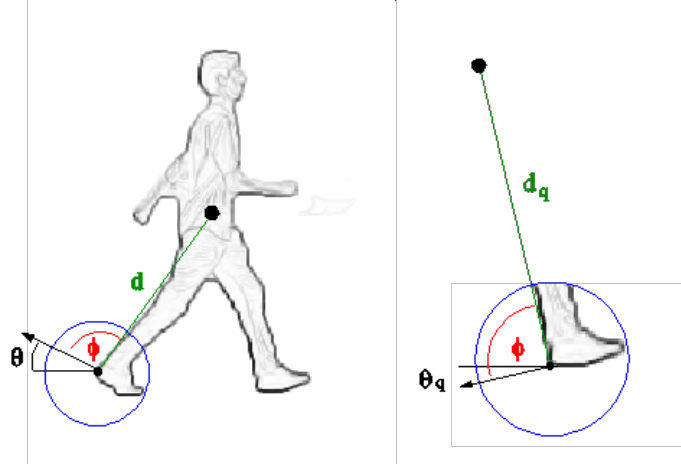


Figure 3.1: Example figure and caption.

Table 3.1: Make sure to set caption to large size.

|      |         |
|------|---------|
| Test | Test    |
| More | Testing |
| More | Testing |

### 3.3 Third Section

This sections hows how to do equations and to show that they number correctly as LaTeX can do with the chapter number and equation number to refer to equation in a piece of text:

$$y = x^2 \tag{3.1}$$

but note it is usually after putting in an equation that you may refer back to equation (3.1) later in the document or chapter.



## 4 TECHNICAL CHAPTER 2

You need to work out how many technical chapters you will require, possibly more than two in this template. Normally your final chapter before the conclusion will include some clear results or proof of concepts from the end goals of your dissertation and it is important to document these well and show the value of what you have achieved. Frequently many dissertations fail to document this information correctly because it is rushed at the end so do ensure you allow time to write it properly.

### 4.1 Section

Text to insert

### 4.2 Section

Text to insert – but ensure the last section really does show the value of your work and where you have got to.

## 5 CONCLUSIONS

Your conclusion should summarise what you have carried out in this work to date, what you have achieved and what you are on course to achieve. Also it should explain what was documented in this dissertation. Note that someone may read this chapter without bothering to read the earlier chapters so there is no problem with repeating what you may have already written earlier. Indeed no new information should be in this part of the conclusion. It should reference back to what is already there.

### 5.1 Evaluation

Stand back and evaluate what you have achieved and how well you have met the objectives. Evaluate your achievements against your objectives in section 1.2. Demonstrate that you have tackled the project in a professional manner.

### 5.2 Future Work

Discuss what further work could be carried out from this work and relate it to what has been carried out.

## BIBLIOGRAPHY

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- [4] Wertz, J. R. and Larson, W. J. . Space Mission Analysis and Design. Space Technology Library. Springer Netherlands, 1999. ISBN 9780792359012.

## APPENDIX

You may have one or more appendices containing detail, bulky or reference material that is relevant though supplementary to the main text: perhaps additional specifications, tables or diagrams that would distract the reader if placed in the main part of the dissertation. Make sure that you place appropriate cross-references in the main text to direct the reader to the relevant appendices. Again you should discuss with your supervisor what appendices are appropriate for this work.

Note that you should not include your program listings as an appendix or appendices. You should submit one copy of such bulky text as a separate item, perhaps on a disk.